Confidence Intervals and Hypothesis Tests

Quick reference

Quantity	Meaning	Main use
p	Unknown population proportion	Parameter of interest
$\hat{p}$	Observed sample proportion	Point estimate; CI standard error
p_0	Null value	Hypothesis-test sampling distribution with mean and standard error
z*	Confidence multiplier	Margin of error
z	Standardized test statistic	Distance from p_0 in null-SE units
P-value	Probability of data at least as extreme under H_0	Evidence against H_0

Core formulas

$$\text{Confidence interval: } \hat{p} \pm z^* \sqrt{[\hat{p}(1 - \hat{p}) / n]}$$

$$\text{One-proportion test: } z = (\hat{p} - p_0) / \sqrt{[p_0(1 - p_0) / n]}$$

Important distinction

For a confidence interval, the standard error uses $\hat{p}$.

For a hypothesis test, the null standard error uses p_0.

In a symmetric two-sided test, double the one-tail probability—not the standard error.

Practice 1 · 95% confidence interval

A random sample of 250 students finds that 145 support extending the campus night-shuttle schedule.

The university enrolls more than 2,500 students. Use $z^* = 1.96$.

1A. Define the population parameter p in words.

1B. Check the conditions. Address random selection, independence / 10% condition, successes, and failures.

1C. Compute $\hat{p}$ and its standard error. Label each value.

1D. Compute the margin of error and the two endpoints.

1E. Interpret the interval in context. State the population, parameter, and confidence level.

Partner check

Does your interval contain $\hat{p}$? Are both endpoints plausible proportions? Did your sentence refer to all students rather than only the sample?

Practice 2 · Confidence, precision, and working backward

Continue with n = 250, $\hat{p}$ = 0.580, and SE = 0.0312.

Part A · A 99% interval

2A-1. Before calculating: should a 99% interval be wider or narrower than a 95% interval from the same data? Explain why.

2A-2. Use z* = 2.576 to find the 99% margin of error and interval.

Part B · Work backward

A reported 95% confidence interval is (0.46, 0.54).

2B-1. Find the center of the interval.

2B-2. Find the margin of error.

2B-3. Use z* = 1.96 to find the standard error.

Comparison sentence

Complete: Higher confidence produces a _____ interval because the critical value z* becomes _____.

Practice 3A · Set up a two-sided hypothesis test

A random sample of 200 students finds that 116 support a proposed advising plan. Test whether the true support proportion differs from 0.50.

3A-1. Define p in words.

3A-2. Write H_0 and H_a. Then explain in words what each says.

3A-3. Explain why this is a two-sided test.

3A-4. Check the conditions under H_0. For the success-failure condition, use $p_0 = 0.50$.

3A-5. Compute $\hat{p}$ and the null standard error SE_0. Why does SE_0 use p_0 rather than $\hat{p}$?

Pause before calculating z

The hypothesis test asks: If p really equaled 0.50, how unusual would a sample proportion of 0.580 be?

Practice 3B · Measure evidence and conclude

Continue the advising-plan test. Use $\hat{p} = 0.580$, $SE_0 = 0.0354$, and $P(Z \geq 2.26) = 0.0119$.

3B-1. Compute the test statistic z. Interpret the sign and magnitude in one sentence.

3B-2. Find the two-sided P-value. Explain why a factor of 2 is used.

3B-3. At $\alpha = 0.05$, state the decision: reject H_0 or fail to reject H_0. Justify it numerically.

3B-4. Write the conclusion in context. Avoid saying that the test proves either hypothesis.

3B-5. Explain what the P-value means using the phrase “at least as extreme.” Then state one thing it does not mean.

CI connection

Without redoing every step, predict whether a 95% confidence interval from these data would include 0.50.
Explain how you know.

Practice 4 · Significance and decision errors

A survey of 20,000 voters estimates support at $\hat{p} = 0.508$. Testing H_0: $p = 0.500$ gives P-value = 0.024.

4A. Is the result statistically significant at $\alpha = 0.05$? Explain.

4B. How large is the estimated difference from 0.500? Express it in percentage points.

4C. Is that difference necessarily important in practice? What additional context or data-quality information would you want?

Decision errors

Suppose H_0 says a safety message does not change helmet use, and H_a says it changes helmet use.

4D. Describe a Type I error in context.

4E. Describe a Type II error in context.

Exit ticket

1. The population parameter in one example today is ...

2. The interval or P-value tells us ...

3. The conclusion we can support is ..., but we still do not know ...

4. Circle one step to practice again: conditions · SE · margin of error · P-value · conclusion